

DS Unit 4 Previous Year Questions

Data Structures (DS) · Unit 4 · Graphs & Graph Algorithms · 2023

Questions collected from previous year university examinations (2019–2023).

Part A — Short answer (2 marks)

1. Explain Graph Representations with a neat diagram and a suitable example. [2021]
2. Compare Breadth-First Search with Depth-First Search. When would you prefer one over the other? [2019]
3. Write short notes on Depth-First Search. [2021]
4. What is meant by Minimum Spanning Trees? Discuss its advantages and limitations. [2022]

Part B — Long answer (10 marks)

1. Explain Dijkstra's Algorithm with a neat diagram and a suitable example. [2022]
2. Illustrate Topological Sorting with a worked example. [2019]
3. Explain Graph Representations with a neat diagram and a suitable example. [2021]
4. Illustrate Breadth-First Search with a worked example. [2020]
5. Write short notes on Depth-First Search. [2020]
6. Compare Minimum Spanning Trees with Topological Sorting. When would you prefer one over the other? [2022]
7. Explain Dijkstra's Algorithm with a neat diagram and a suitable example. [2023]
8. Discuss the role of Topological Sorting in Graphs & Graph Algorithms. [2022]
9. Compare Graph Representations with Topological Sorting. When would you prefer one over the other? [2023]
10. Illustrate Breadth-First Search with a worked example. [2021]